



Republic of Rwanda
Ministry of Education



RTB | RWANDA
TVET BOARD

SWDDD401

DATABASE DEVELOPMENT

Develop Database

Competence

RQF Level: 4

Learning Hours



100

Credits: 10

Sector: ICT and Multimedia

Trade: Software Development

Module Type: Specific

Curriculum: ICTSWD4002 TVET Certificate IV in Software Development

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1200

Issue Date: September 2023

Purpose statement	This module describes the knowledge, skills and attitudes to be acquired by the learner to Analyse database, design database, implement database and secure database. At the end of this module the learner will be able to analyse database, design database, implement database and secure database.					
Learning assumed to be in place	N/A					
Delivery modality	Training delivery		100%	Assessment		Total 100%
	Theoretical content		30%	Formative assessment	30%	50%
	Practical work:		70%		70%	
	Group project and presentation	20%				
	Individual project /Work	50%				
			Summative Assessment		50%	

Elements of Competence and Performance Criteria

Elements of competence	Performance criteria
1. Analyse Database	1.1 Database fundamentals are properly described based on database standards
	1.2 Data dictionary is clearly described based on database model
	1.3 Database Task requirements are properly identified based on user requirements
2. Design Database	2.1 Conceptual Database schema is properly designed based on system requirements
	2.2 Logical Database schema is properly designed based on system requirements
	2.3 Database optimization is effectively enforced based on database schema
	2.4 Physical Database Schema is appropriately created based on the Physical Data Model.
3. Implement Database	3.1 Data definition language commands are effectively applied based on database schema
	3.2 Data manipulation language commands are effectively applied based on database schema
	3.3 Data control language commands are effectively applied based on database schema
	3.4 Data Query Language commands are effectively applied based on database schema
	3.5 Transaction Control Language commands are effectively applied based on database schema

4. Secure Database	4.1 Access control is properly enforced based on database security measures
	4.2 Auditing and logging are clearly managed based on the security policies
	4.3 Data encryption is correctly implemented based on data security measures
	4.4 Backup and Recovery of data are regularly configured based on DBMS

Intended Knowledge Skills and Attitudes

Knowledge	Skills	Attitudes
✓ Describe database fundamentals ✓ Describe data dictionary ✓ Identify database requirements ✓ Describe database schema ✓ Describe SQL	✓ Design conceptual database schema ✓ Design logical database schema ✓ Optimize database ✓ Design Physical database schema ✓ Apply DDL, DML, DQL, DCL and TCL commands ✓ Enforce data access control ✓ Manage Auditing and logging ✓ Implement Data encryption ✓ Configure database backup and restore	✓ Creativity ✓ Innovation ✓ Adaptability ✓ Flexibility ✓ Teamwork ✓ Confidence ✓ Time management ✓ Passion ✓ Integrity ✓ Curiosity ✓ Organized

Course content

Learning outcomes	At the end of the module the learner will be able to: 1. Analyse Database 2. Design Database 3. Implement Database 4. Secure Database
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Learning outcome 1: Analyse Database	Learning hours: 20
Indicative content	
• Description of database fundamental ✓ Definition of key terms - Database - Data - Information - Entities - Attributes/Field - Records - Table - Database schema - DBMS - SQL ✓ Applications of database ✓ Advantages and Disadvantages of Database ✓ Identification of database models - Relational database - Hierarchical database - Network database	

✚ Object oriented model ✓ Identification of database Relationship ✚ One to one ✚ One to many ✚ Many to one ✚ Many to many ✓ Determination of data types ✚ Character ✚ Number ✚ Date • Description of data dictionary ✓ Definition of data dictionary ✓ Elements of data dictionary • Identification of database requirements ✓ Types of database requirements ✚ Functional requirement ✚ Non-functional requirement ✓ Methods to collect data ✚ Interview ✚ Documentation ✚ Questionnaire ✚ Observation	
Resources required for the learning outcome	
Equipment	▪ Computer
Materials	▪ Internet
Tools	▪ NA

Facilitation techniques	▪ Brainstorming ▪ Demonstration ▪ Group discussion practical work
Formative assessment methods /(CAT)	▪ Written assessment ▪ Oral presentation ▪ Performance assessment

Learning outcome 2: Design Database	Learning hours: 25
Indicative content	
• Description of database schema ✓ Introduction of database schema ✓ Types of database schema ✓ Data abstraction levels ✓ Types of data independence • Design of conceptual database schema ✓ Description of conceptual database schema ✓ Entity relationship diagram (ERD) ✚ Description of ERD ✚ Components of ERD ✚ Define relationships ✚ Create an ERD ✚ Draw an ERD (MS-Visio, Draw-Max) • Design of logical database schema ✓ Description of logical database schema ✓ Table constraints ✚ NOT NULL Constraint. ✚ UNIQUE Constraint.	

-  DEFAULT Constraint.
-  CHECK Constraint.
-  PRIMARY KEY Constraint.
-  FOREIGN KEY Constraint.

✓ Convert conceptual database schema to logical database schema

- **Optimization of database**

✓ Data normalization

 First normal form (1NF)

 Second normal form (2NF)

 Third normal form (3NF)

✓ Indexing

- **Design of Physical database schema**

✓ Description of DBMS

✓ Preparation of DBMS Environment (MySQL)

✓ Convert logic database schema to physical database schema

Resources required for the learning outcome

Equipment	▪ Computer
Materials	▪ Internet
Tools	▪ E-Draw max ▪ Microsoft Visio ▪ MySQL
Facilitation techniques	▪ Brainstorming ▪ Demonstration ▪ Group discussion practical work
Formative assessment methods /(CAT)	▪ Written assessment ▪ Oral presentation ▪ Performance assessment

- Project based assessment

Learning outcome 3: Implement Database

Learning hours: 36

Indicative content

- **Description to SQL**
 - ✓ Introduction of SQL
 - ✓ SQL sub-languages
 - ✓ SQL Operators
 - ✚ SQL Arithmetic Operators
 - ✚ SQL Bitwise Operators
 - ✚ SQL Compound Operators
 - ✚ SQL Logical Operators
- **Application of DDL commands**
 - ✓ CREATE
 - ✚ Database
 - ✚ Table Constraints
 - ✚ Table
 - ✓ ALTER Table
 - ✓ DROP
 - ✚ Database
 - ✚ Table
 - ✓ TRUNCATE Table
 - ✓ MODIFY
 - ✚ Database
 - ✚ Table
- **Application of DML commands**
 - ✓ INSERT

- ✓ UPDATE
- ✓ DELETE
- ✓ CALL
- ✓ EXPLAIN CALL
- ✓ LOCK
- **Application of DQL Command**
 - ✓ SELECT
 - ✓ SQL aggregate function
 - ✓ SQL clause
- **Application of DCL commands**
 - ✓ GRANT
 - ✓ REVOKE
- **Application of TCL commands**
 - ✓ COMMIT
 - ✓ SAVEPOINT
 - ✓ ROLLBACK
 - ✓ SET Transaction
 - ✓ SET Constraints

Resources required for the learning outcome

Equipment	▪ Computer
Materials	▪ Internet
Tools	▪ MySQL ▪ Apache ▪ XAMPP ▪ WAMP ▪ LAMP

	▪ Browsers
Facilitation techniques	▪ Brainstorming ▪ Demonstration ▪ Group discussion practical work
Formative assessment methods /(CAT)	▪ Written assessment ▪ Oral presentation ▪ Performance assessment ▪ Project based assessment

Learning outcome 4: Implement Database security	Learning hours: 19
Indicative content	
• Enforcement of data access control ✓ Description of database security + Introduction of database security + Types of database security ✓ Data access control ✓ Access control policies + Identify the data classifications + Define roles and permission ✓ Authentication + Identify user accounts + Create privileges + Configure the authentication system + Test the authentication system + Monitor and maintain ✓ Authorization + Create roles	

- + Assign permissions/privilege to roles
- + Assign roles to users
- + Test the authorisation system
- + Monitor and maintain

- **Management of Auditing and logging**

- ✓ Logging

- + Identify the logging requirements
 - + Configure logging settings
 - + Monitor log data
 - + Analyse log data
 - + Archive log data
 - + Corrective action

- ✓ Auditing

- + Identify the data that needs to be audited
 - + Execution of SQL command
 - + Configure audit settings
 - + Review audit
 - + Analyse audit data
 - + Corrective action

- **Implementation of Data encryption**

- ✓ Description of data encryption

- ✓ Application of encryption technics

- + Symmetric Encryption
 - + Asymmetric Encryption
 - + Hashing

- **Configuration of database backup and restore**

- ✓ Introduction of data backup and restore

- ✓ Backup Method

 Full backup  Differential backup  Incremental backup ✓ Backup schedule ✓ Create Backup ✓ Perform recovery method  Full database recovery  Rollback recovery  Point-in-time recovery ✓ Test your backup and recovery plan	
Resources required for the learning outcome	
Equipment	▪ Computer
Materials	▪ Internet
Tools	▪ MySQL ▪ Apache ▪ XAMPP ▪ WAMP ▪ LAMP ▪ Browsers
Facilitation techniques	▪ Brainstorming ▪ Demonstration ▪ Group discussion practical work
Formative assessment methods /(CAT)	▪ Written assessment ▪ Oral presentation ▪ Performance assessment ▪ Project based assessment

Integrated/Summative assessment

Integrated situation

D-STORE Ltd is a company that generates revenue from selling furniture products; It's located in Nyabihu district, western province.

D-STORE Ltd uses file system (books) to record information about the sales and inventory.

Company has a problem of non-efficient security, accessibility and management of information about the stock-in, stock-out and customers. D-STORE Ltd company wants to switch to a digital system that can be accessed easily by authorized employees.

They hired a database developer who develops for them the database that can be used to manage sales, inventory, customer data, and to generate an invoice for all any placed order.

Tasks to be performed:

1. Analyze the database requirements and identify the different data sources that need to be integrated.
2. Based on the requirements, design a database that includes conceptual database schema and logic database schema. also prepare a MySQL DBMS environment ^[1]_{SEP}
3. In this stage, create the physical database schema by applying SQL commands in DBMS and Developer must develop a query to insert customer order with the following information: Customer information and product information. And make another query to retrieve data to be on invoicing such as: Customer information and product information, after that a developer also must generate a product status report.
4. Perform database access control: apply database user authentication, database authorization user and audit logs

5. Configuration of database backup

Instruction

1. Database is needed in 6 hours
2. Build the database with respect of table constraints
3. No Data redundancy allowed in database
4. All query performed must documented
5. D-STORE Ltd has the following employees' positions:
 - Managing director: They manage all company activities
 - Stock manager: Manage and control all product in and out of the stock
 - Stocker worker: Record sold products

Tools	▪ MySQL, Apache, XAMPP, WAMP, LAMP, Browsers, E-Draw max
Equipment	▪ Computer
Materials/ Consumables	▪ Internet

Assessable outcomes	Assessment criteria (Based on performance criteria)	Indicator	Observation		Marks allocation
			Yes	No	
Learning outcome1: Analyse database: (10%)	1.Gathering requirements	Ind.1: Database requirement documentation is papered			5
Learning Outcome2: Database design:	1.Conceptual and logical database schema design	Ind.1: Entities are drown			3
		Ind.2: Attributes are drown			2

(30%)		Ind.3: Primary key are shown			1
		Ind.4: Foreign key are shown			1
		Ind.5: Relationship are identified			2
		Ind.6: logic database schema is sketched			2
	2.Prepare DBMS Environment	Ind.1: XAMPP is installed			2
		Ind.2: Apache is started			1
		Ind.3: MySQL is started			1
Learning Outcome3: Implement database: (40%)	1.Apply SQL Commands	Ind.1: Database is created			2
		Ind.2: Tables are created			2
		Ind.3: Relationship is created			2
		Ind.4: Data are inserted			2
		Ind.5: Product Order is performed			4
		Ind.6: Invoice data are Fetched			4
		Ind.7: Payment status are updated			2

		Ind.8: Product quantities are updated			2
Learning Outcome 4: Secure database: (20%)	1.Apply database access control	Ind.1: Database users are created			2
		Ind.2: Roles are created			1
		Ind.3: Privileges are assigned to roles			2
		Ind.4: Roles are assigned to users			2
		Ind.5: Logs are defined			1
	2.Configure backup and restore	Ind.1: Backup is created			2
Total marks		50			
Percentage Weightage		100%			
Minimum Passing line % (Aggregate): 70%					

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